Lecture 20

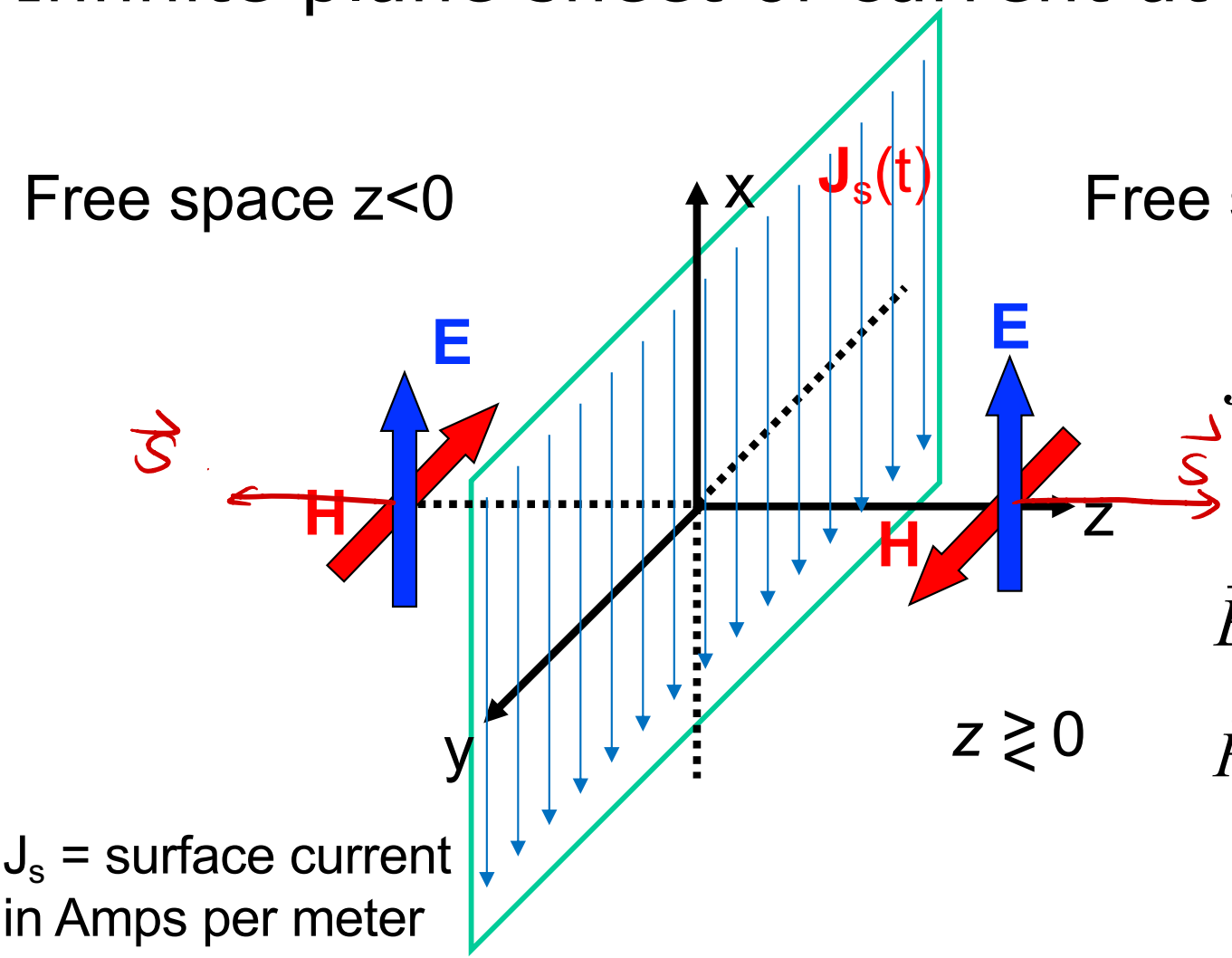
Power Flow/Energy Storage
Poynting's Theorem

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Field Source

- Infinite plane sheet of current at z=0



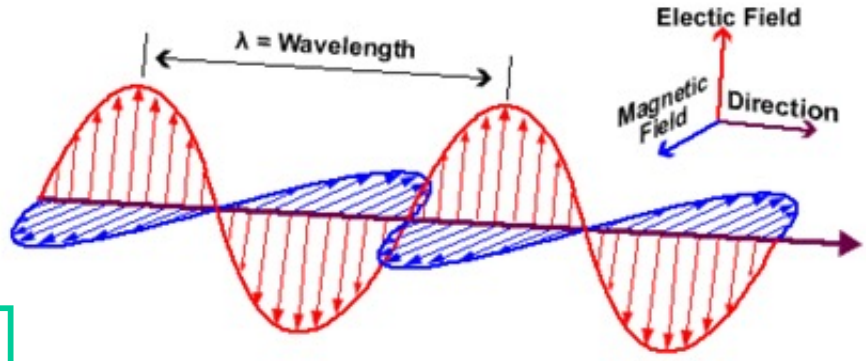
$$\vec{J}_S = -J_x(t)\hat{a}_x$$

At z=0

$$\vec{E} = E_x(z,t)\hat{a}_x$$

$$\vec{H} = H_y(z,t)\hat{a}_y$$

Combining the expressions…

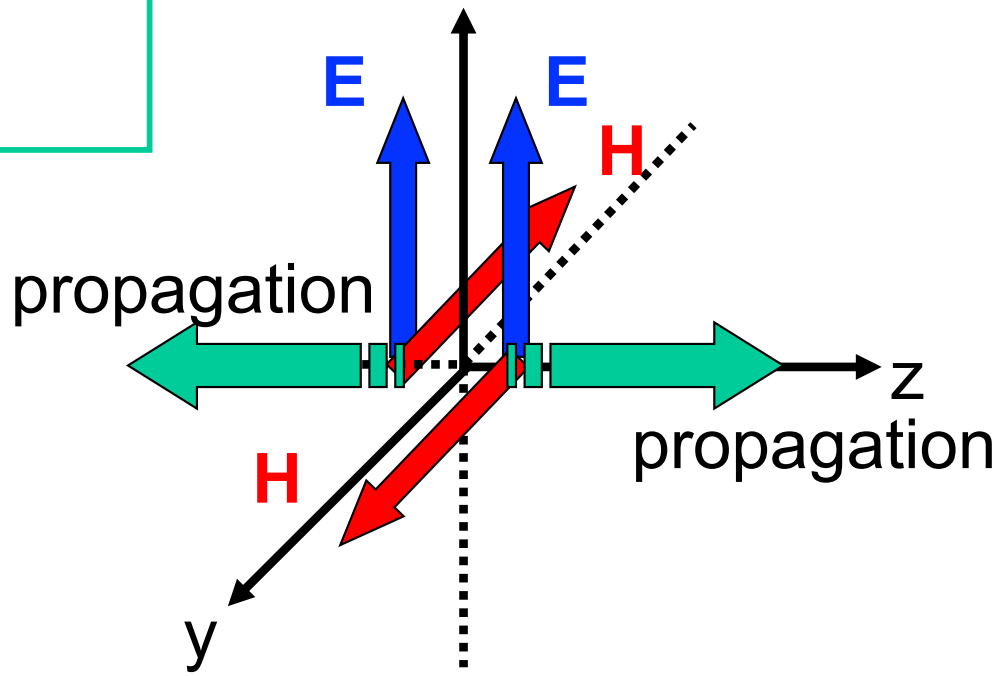


$120\pi\,\Omega$

$|E| = \eta_0 |H|$

$$\vec{E}(z,t) = \frac{\eta_0}{2} J_S\left(t \mp \frac{z}{v_p}\right)\hat{a}_x \qquad z \gtrless 0$$

$$\vec{H}(z,t) = \pm\frac{1}{2} J_S\left(t \mp \frac{z}{v_p}\right)\hat{a}_y$$



Definition: Poynting Vector

The POWER provided by the current source is used to generate the propagating **E** and **H** fields.

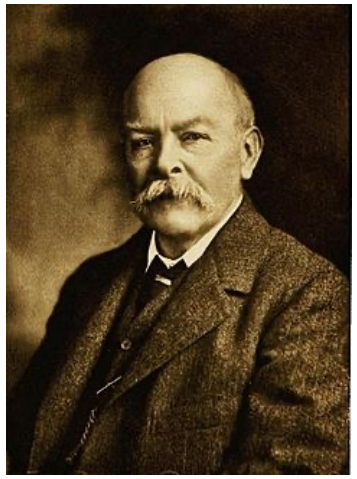
The **E** and **H** fields are carrying power with them as they propagate

$$\vec{S} = \vec{E} \times \vec{H}$$

$\frac{V}{m}$ $\frac{A}{m}$.

Power Flow Density of an EM Field

Units for **S**: Watts/m^2



John H. Poynting
(1852–1914)

Integral & Differential Forms

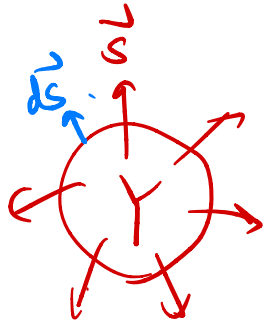
$$\vec{S} = \vec{E} \times \vec{H}$$

$\vec{S}(z,t)$

"Instantaneous" Poynting vector

We can calculate power density magnitude and direction for any single place and time if we know **E** and **H** at that place and time.

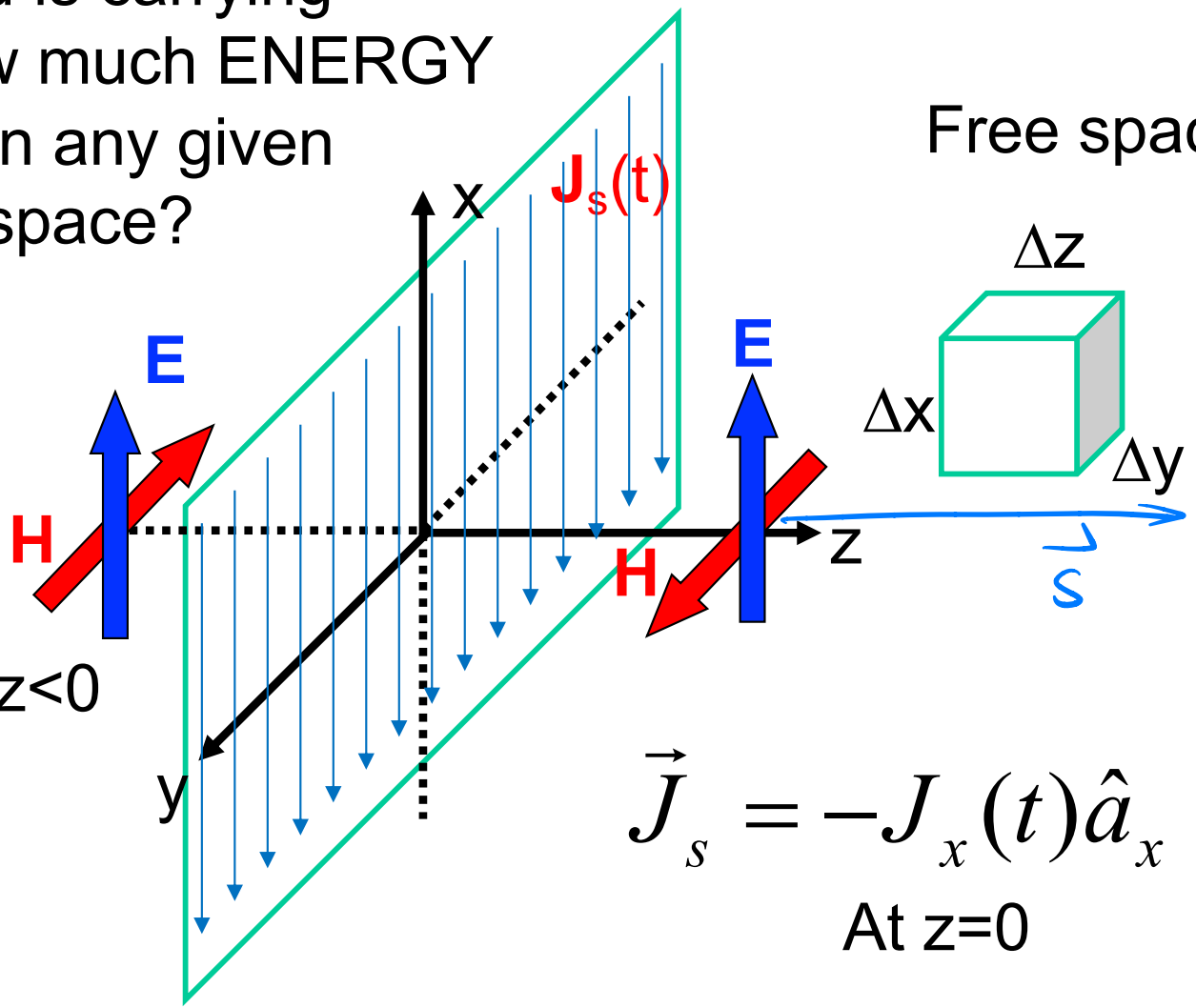
$$\oiint_S \vec{S} \bullet d\vec{S} = \oiint_S \left(\vec{E} \times \vec{H}\right) \bullet d\vec{S}$$



Power flow **<u>out</u>** of a CLOSED surface (units = Watts)

There's ENERGY in the field!

If the EM field is carrying POWER, how much ENERGY is contained in any given VOLUME of space?



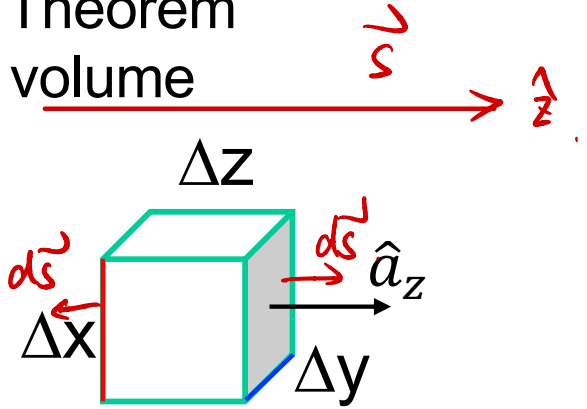
$$\vec{J}_s = -J_x(t)\hat{a}_x$$

At z=0

How much ENERGY?

$$\vec{S} = \vec{E} \times \vec{H} = |E_x||H_y|\hat{a}_z$$

Use integral form to get a special case of Poynting's Theorem and calculate the power flow OUT of our little closed volume



$$\oiint_S \vec{S} \cdot d\vec{S} = S_z\big|_{z+\Delta z} \cdot \Delta x \Delta y - S_z\big|_{z} \cdot \Delta x \Delta y$$

(right: $z+\Delta z$; left: z)

$$= \left(\frac{S_z|_{z+\Delta z} - S_z}{\Delta z} \right) \cdot \Delta x \Delta y \Delta z = \frac{\partial S}{\partial z} \cdot \Delta V$$

(ΔV: volume)

How much energy is "stored" in **E** and **H** for this volume ΔV?

$S_z = E_x H_y$

$$\oiint_S \vec{S} \cdot d\vec{S} = \frac{\partial [E_x H_y]}{\partial z} \Delta V = \left(H_y \frac{\partial E_x}{\partial z} + E_x \frac{\partial H_y}{\partial z} \right) \Delta V$$

$$= \left(-\mu_0 H_y \frac{\partial H_y}{\partial t} - \varepsilon_0 E_x \frac{\partial E_x}{\partial t} \right) \Delta V$$

$$= -\frac{\partial}{\partial t}\left(\frac{1}{2}\mu_0 H_y^2 \Delta V \right) - \frac{\partial}{\partial t}\left(\frac{1}{2}\varepsilon_0 E_x^2 \Delta V \right)$$

H energy / volume

E Energy / volume

Hint: Use coupled 1st order PDE for plane wave propagation (Lecture 18, slide 13)

$$\frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t} = -\mu_0 \frac{\partial H_y}{\partial t}$$

$$\frac{\partial H_y}{\partial z} = -J_x - \frac{\partial D_x}{\partial t} = -\epsilon_0 \frac{\partial E_x}{\partial t}$$

J_x ↑ $z=0$

$z \gtrless 0$

$J_s = J_{s0}\cos(\omega t)\,(-\hat{x})$

$\vec{E} = \frac{J_{s0}}{2}\eta_0 \cos(\omega t \mp \beta z)\,\hat{x}$

$\vec{H} = \pm \frac{J_{s0}}{2}\cos(\omega t \mp \beta z)\,\hat{y}$ $\quad H_0$

$\beta = \frac{2\pi}{\lambda} = \frac{\omega}{v_p}$

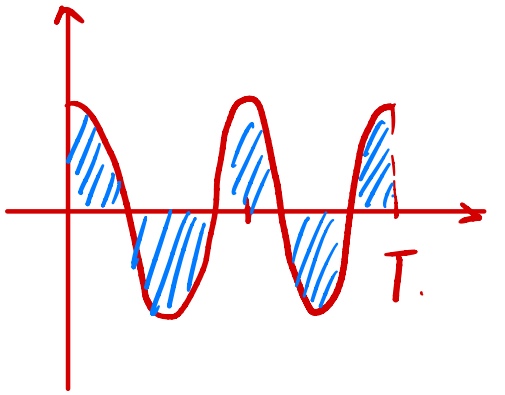
A question we may ask on an exam

The power is not a constant value as a function of time.
Remember - it has that $\cos^2(t)$ dependence.

What is the TIME-AVERAGED power being carried by the EM field?

$$\langle S_z \rangle = \frac{1}{T}\int_0^T S_z(t)\,dt$$

where period $T = 1/f$



$$\langle S_z \rangle = \left\langle \frac{\eta_0 J_{S0}^2}{4}\cos^2(\omega t - \beta z) \right\rangle$$

$$= \frac{\eta_0 J_{s0}^2}{4} < \frac{1+\cos(2(\omega t-\beta z))}{2} > \quad 0$$

$$= \frac{\eta_0 J_{s0}^2}{8} \quad \text{const.} \quad \text{no } z, / \text{no } t.$$

$$= \frac{\eta_0}{2} H_0^2 \quad [\,W/m^2\,]$$

$\vec{E} = \eta_0 H_0 \cos(\omega t - \beta z)\hat{x}$

Power calculations

For $\vec{H} = H_0 \cos(6\pi\times10^7 t - 0.2\pi z)\,\hat{a}_y$ A/m, find: $S: +\hat{z}$.

(a) the instantaneous power flow across a 1 m² area, A, in the z=0 plane at t=0

$\iint \vec{S}\cdot d\vec{S} = \eta_0 H_0^2 \cos^2(\omega t - \beta z)\cdot 1\,m^2 \cdot \hat{z}$.

$t=0,\ z=0$

$P = \eta_0 H_0^2$ [W]

(b) the instantaneous power across A in the z=0 plane at $t = 1/8\ \mu s$

$P = \eta_0 H_0^2 \cos^2\left(6\pi\times10^7 \times \frac{1}{8}\times10^{-6} - 0\right) = \eta_0 H_0^2 \cos\left(\frac{15}{2}\pi\right) = 0$.

$\cos\left(\frac{3}{2}\pi\right)$

(c) the time averaged power flow across A

$P_{avg} = \frac{1}{2}\eta_0 H_0^2 = 60\pi H_0^2$.

Poynting's Theorem

Begin with: $\nabla \bullet \vec{S} = \nabla \bullet (\vec{E} \times \vec{H}) = \vec{H} \bullet (\nabla \times \vec{E}) - \vec{E} \bullet (\nabla \times \vec{H})$

With Maxwell's curl equations

$$\nabla \times \vec{\boldsymbol{E}} = -\frac{\partial \vec{\boldsymbol{B}}}{\partial t} = -\mu_0 \frac{\partial \vec{\boldsymbol{H}}}{\partial t} \qquad \nabla \times \vec{\boldsymbol{H}} = \vec{\boldsymbol{J}} + \frac{\partial \vec{\boldsymbol{D}}}{\partial t} = \vec{\boldsymbol{J}} + \epsilon_0 \frac{\partial \vec{\boldsymbol{E}}}{\partial t}$$

$$\nabla \cdot (\vec{\boldsymbol{E}} \times \vec{\boldsymbol{H}}) = \vec{\boldsymbol{H}} \cdot \left(-\mu_0 \frac{\partial \vec{\boldsymbol{H}}}{\partial t}\right) - \vec{\boldsymbol{E}} \cdot \left(\vec{\boldsymbol{J}} + \epsilon_0 \frac{\partial \vec{\boldsymbol{E}}}{\partial t}\right)$$

$\nabla \times \vec{E}$ $\qquad$ $\nabla \times \vec{H}$

Poynting's Theorem

Thus, $$\nabla \bullet \vec{S} = -\frac{\partial}{\partial t}\left(\frac{1}{2}\varepsilon_0 E^2\right) - \frac{\partial}{\partial t}\left(\frac{1}{2}\mu_0 H^2\right) - \vec{E} \bullet \vec{J}$$

$$\nabla \bullet \vec{S} + \vec{E} \bullet \vec{J} = -\frac{\partial w_e}{\partial t} - \frac{\partial w_m}{\partial t}$$

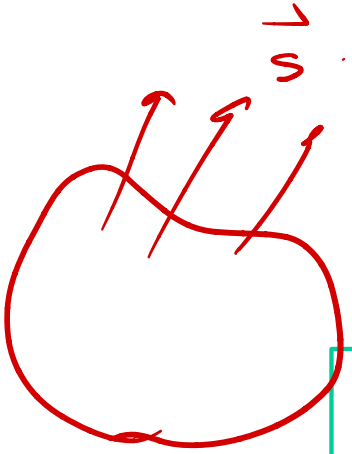
$$-\frac{\partial}{\partial t}(w_e + w_m) = \nabla \bullet \vec{S} + \vec{E} \bullet \vec{J}$$

$$w_e = \frac{1}{2}\epsilon_0 E^2, w_m = \frac{1}{2}\mu_0 H^2$$

w_e = electric field energy density
w_m = magnetic field energy density

Poynting's Theorem

$$\nabla \cdot \vec{S} + \frac{\partial}{\partial t}\left(\frac{1}{2}\epsilon_0 \vec{\boldsymbol{E}} \cdot \vec{\boldsymbol{E}} + \frac{1}{2}\mu_0 \vec{\boldsymbol{H}} \cdot \vec{\boldsymbol{H}}\right) + \vec{E} \cdot \vec{J} = 0$$



Integrate over the volume and Apply Divergence Theorem:

$$\oiint_S \vec{S} \bullet d\vec{S} = \iiint_V \nabla \bullet \vec{S}\ dV$$

$$\oiint_S \vec{S} \cdot d\vec{S} + \frac{\partial}{\partial t}\iiint_V (w_e + w_m) dV + \iiint_V \vec{E} \cdot \vec{J}\, dV = 0$$

Power flow **OUT** of surface + Time rate of energy stored in E and H field + Joule heating (Work done by the field) =0

Can be + or -

Can be + or -

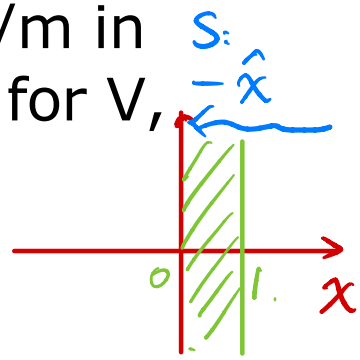
- $\vec{E} \cdot \vec{J} > 0$ for resistive load. J=σE, so $\vec{E} \cdot \vec{J}$ = σ|E|^2 ≥ 0
- $\vec{E} \cdot \vec{J}$ =0 only if σ=0 (perfect dielectric)
- $\vec{E} \cdot \vec{J}$<0 when J is a source of EM energy.

z=0 . J : $-\hat{x}$

E : $+\hat{x}$

Challenge Question: Poynting's Theorem

- If $\vec{\boldsymbol{H}} = H_0 \cos(ct + x)\, \hat{\boldsymbol{a}}_y$ A/m and $\vec{\boldsymbol{E}} = \eta_0 H_0 \cos(ct + x) \hat{\boldsymbol{a}}_z$ V/m in the *free space* region x≥0, which statement is **true** for V, the volume bounded by the x=0 and x=1 planes



(a) the net outward power flow is zero at all times

(b) the fields do work and thus lose energy

(c) the total electric field energy inside is constant in time

(d) the time averaged electric field energy density at each position x is the same

(e) the total electric and magnetic field energy density at fixed position x is constant in time

Lecture 20 Summary

- Poynting's Theorem:
  - Power into a volume from electromagnetic fields is equal to the power converted into heat in the volume plus the increase in the energy stored in the fields in the volume

$$-\oiint_S \vec{S}\cdot d\vec{S} = \frac{\partial}{\partial t}\iiint_V \left(\frac{1}{2}\epsilon_0 E^2 + \frac{1}{2}\mu_0 H^2\right) dV + \iiint_V \vec{E}\cdot\vec{J}\, dV$$

  - Time averaged Poynting vector

$$\left\langle \vec{S} \right\rangle = \frac{1}{2}\mathrm{Re}\left[\widetilde{E}\times\widetilde{H}^*\right]$$

- Next up: Waves in materials